

Alex Short

alex.abe.short@gmail.com | ashort04@calpoly.edu | shorale21.github.io

Education

California Polytechnic State University, San Luis Obispo (Since Fall 2021)		Estimated Completion:
Major:	Bachelor of Science in Physics	June 2025
2 nd Major:	Bachelor of Science in Computer Engineering	June 2025
Minor:	Mathematics	Completed June 2023

Research and Projects

Loop Quantum Gravity | Date: Present | Advisor: Dr. Lei Lu

- Exploring the canonical quantization of **general relativity** using **Yang-Mills** theories and **Chern-Simons** theories to develop a pedagogical publication on the subject of **loop quantum gravity**

AdS/CFT Research | Date: Future | Advisor: Dr. Lei Lu

- I will explore the AdS/CFT dynamics of the magnetic black brane to learn about **holographic renormalization** and how quantum fields can be correlated to higher dimensional **gravitational systems**

Diffusion in Microhydrodynamical Systems | Date: Present | Advisor: Dr. Jon Fernsler

- Determination the effects of fluid flow on microscopic diffusion present in quasi-two dimensional systems (fluid-thermal coupling in viscous fluids)
- Use of **fluorescence microscopy** to observe lipid monolayers in a Langmuir trough setup in order to quantify diffusion
- Exploration of the aggregative properties of quantum dots embedded in a thin smectic liquid crystal film
- Presented at BCOSAM Spring Research Conference 2023 and 2024

SLAC Internship | Date: July - August 2024 | Advisor: Claudio Emma

- Interned with the SULI program at SLAC at **Stanford University**
- Worked on simulations of XTCAV components for the FACET-II Accelerator
- Presented at SLAC intern conference 2024

USB to Automotive-Ethernet Adapter for Zoox | Date: Present | Advisor: Dr. John Oliver

- I am the **Project Manager** of a small team designing a self-contained adapter from USB 3.0 to 1000 BaseT Automotive Ethernet
- Skills: PCB design, circuit design, leadership skills, project organization
- Our client is Zoox, a self driving car company

University of Colorado Boulder REU |Date: July – August 2023 |

Advisor: Dr. Joe MacLennan & Dr. Cheol Park

- Observation of the diffusive properties of smectic C islands in liquid crystals
- Use of polarized microscopy to **observe topological vortices** in liquid crystal films
- Use of artificial intelligence to track and quantify the diffusion of features in liquid crystal films, as well as image processing and statistical analysis of the results
- Presented at CU Boulder Physics Summer Research Conference 2023

Frost SURP: Shape Dynamics & Gravitational Physics | Date: February – August 2022 |

Advisor: Dr. Sean Echols & Dr. Julian Barbour

- Use of simulations to understand **classical gravitational** systems
- Modeling and understanding of ‘shape space’ in the classical sense in order to find **quantum properties of gravitation**
- Presented at FROST Summer Research Conference (Cal Poly) 2022

Relevant Skills & Knowledge

- I am skilled with the following programming languages: C, C++, Mathematica, Matlab, Python, Java, HTML/CSS, Verilog, Systemverilog, RISC-V Assembly, ARM Assembly, x86 Assembly
- Mathematical Skills: Advanced Tensor Calculus, Field Theory, Real Analysis, Linear Algebra, Abstract Algebra, Partial Differential Equations, Differential Geometry
- FPGA Design, Implementation, and Programming
- Microcontroller Design, Implementation, and Programming (Including Arduino, Nano, and STM32)
- Multi-layer PCB design and layout
- Electronic circuit design and digital signal processing
- I am knowledgeable about computer graphics and hardware security

Reading List (Notable Books I Have Read)

- “Loops, Knots, Gauge Theories, and Quantum Gravity” by R. Gambini and J. Pullin
- “A First Course in Loop Quantum Gravity” by R. Gambini and J. Pullin
- “Gauge Fields, Knots, and Gravity” by J. Baez and J. Muniain
- “Spacetime and Geometry” by S. Carroll
- “Classical Mechanics” by J. Taylor
- Selections from “Classical Mechanics” by H. Goldstein, C. Poole, and J. Safko
- “Introduction to Electrodynamics” by D. Griffiths
- “Introduction to Quantum Mechanics” by D. Griffiths
- Selections from “Quantum Mechanics” by N. Zettili
- “Abstract Algebra 7th Edition” by J. Fraleigh
- Selections from “Foundations of Differential Geometry” by Kobayashi & Nomizu
- “Linear Algebra Done Right” by S. Axler
- “Vector Calculus, Linear Algebra, and Differential Forms, a Unified Approach” by J. Hubbard and B. Hubbard
- “Applied Partial Differential Equations” by R. Haberman
- “The Principles of Electromagnetic Theory and of Relativity” by M. Tonnelat
- First 20 chapters from “A Standard Model Workbook” by T. Moore